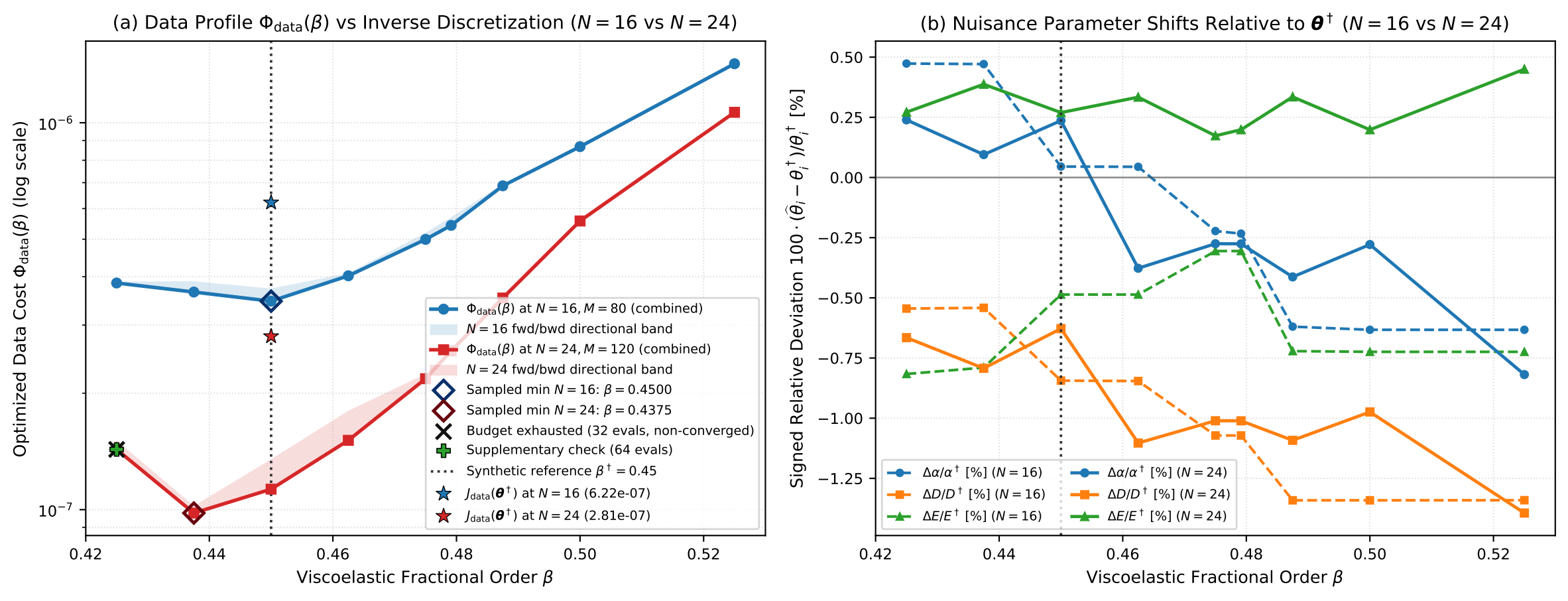
(a) Data Profile $\Phi_{\text{data}}(\beta)$ vs Inverse Discretization ($N = 16$ vs $N = 24$)
Optimized Data Cost $\Phi_{\text{data}}(\beta)$ (log scale)
10^{-6}
10^{-7}
0.42
0.44
0.46
0.48
0.50
0.52
Viscoelastic Fractional Order β
$\Phi_{\text{data}}(\beta)$ at $N = 16, M = 80$ (combined)
$N = 16$ fwd/bwd directional band
$\Phi_{\text{data}}(\beta)$ at $N = 24, M = 120$ (combined)
$N = 24$ fwd/bwd directional band
Sampled min $N = 16$: $\beta = 0.4500$
Sampled min $N = 24$: $\beta = 0.4375$
Budget exhausted (32 evals, non-converged)
Supplementary check (64 evals)
Synthetic reference $\beta^\dagger = 0.45$
$J_{\text{data}}(\boldsymbol{\theta}^\dagger)$ at $N = 16$ (6.22e-07)
$J_{\text{data}}(\boldsymbol{\theta}^\dagger)$ at $N = 24$ (2.81e-07)
(b) Nuisance Parameter Shifts Relative to $\boldsymbol{\theta}^\dagger$ ($N = 16$ vs $N = 24$)
Signed Relative Deviation $100 \cdot (\widehat{\theta}_i - \theta_i^\dagger)/\theta_i^\dagger$ [%]
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
0.42
0.44
0.46
0.48
0.50
0.52
Viscoelastic Fractional Order β
$\Delta\alpha/\alpha^\dagger$ [%] ($N = 16$)
$\Delta D/D^\dagger$ [%] ($N = 16$)
$\Delta E/E^\dagger$ [%] ($N = 16$)
$\Delta\alpha/\alpha^\dagger$ [%] ($N = 24$)
$\Delta D/D^\dagger$ [%] ($N = 24$)
$\Delta E/E^\dagger$ [%] ($N = 24$)